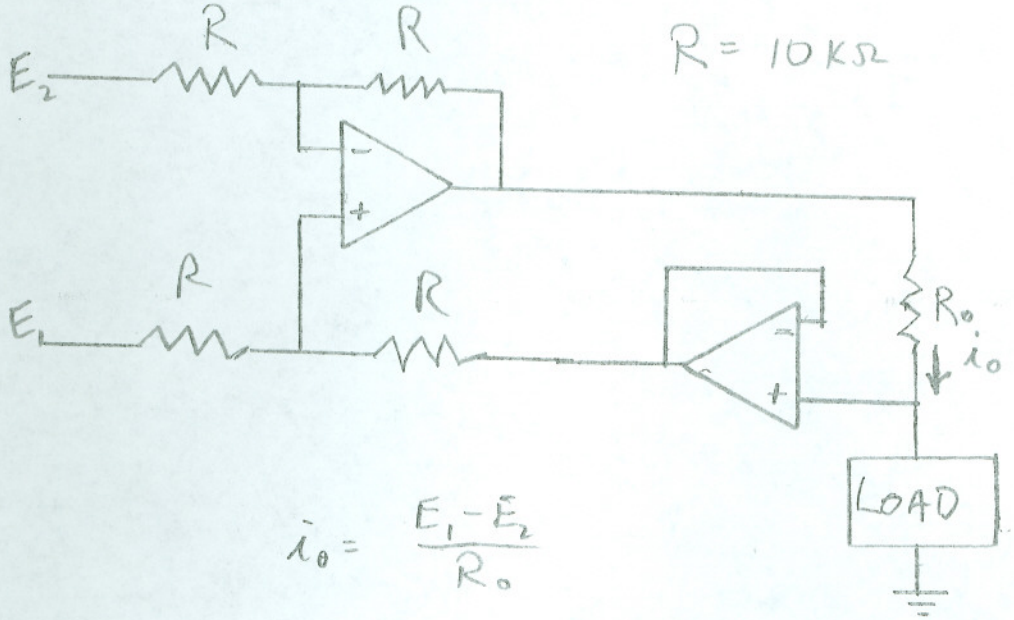
Analytical Cume
March 4, 2005

The development of the integrated circuit version of the operational amplifier revolutionized modern instrumentation about 30 years ago. These devices still find uses in all modern instrumentation.

Attached you will find two circuits based on operational amplifiers. The first circuit is called a differential voltage-to-current converter and the second is an instrument amplifier.

The theoretical output for each circuit is shown. Derive (50 points, each) this output for each circuit. To receive full credit your analysis should include the application of Kirchoffs law, and Ohm's law to the circuit.

①

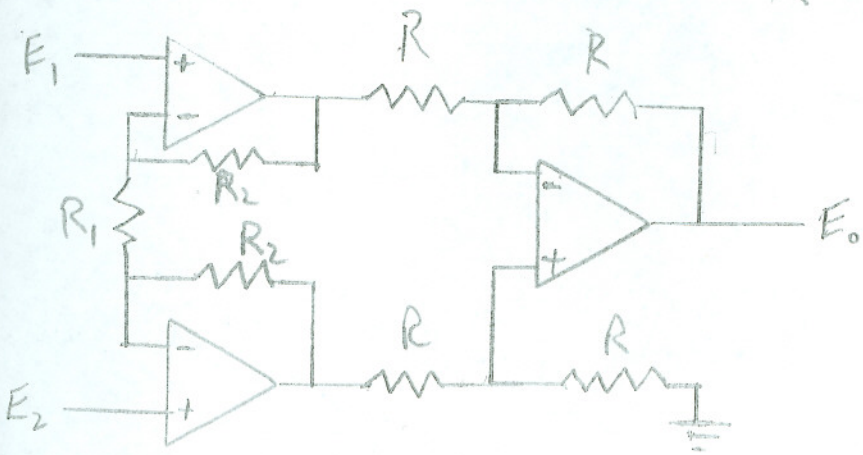


$R = 10\,K\Omega$

$$i_o = \frac{E_1 - E_2}{R_o}$$

②

$R = 10\,K\Omega$



$$E_o = \left(1 + \frac{2R_2}{R_1}\right)(E_2 - E_1)$$